



OPEN Machine learning models for predicting multimorbidity trajectories in middle-aged and elderly adults

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Multimorbidity has emerged as a significant public health issue in the context of global population aging. Predicting and managing the progression of multimorbidity in the elderly population is crucial. This study aims to develop predictive models for multimorbidity trajectories in middle-aged and elderly populations and to identify the key factors influencing the progression of multimorbidity. First, a time-series clustering method was used to construct the multimorbidity trajectories. Then, predictive models based on machine learning techniques were developed to forecast the progression of different trajectories and identify key risk factors. This study utilized data from the China Health and Retirement Longitudinal Study (CHARLS) database, including 12,198 middle-aged and elderly individuals (aged 45 and above). Four distinct multimorbidity progression patterns were identified: Stable Low-Risk Group (45.26%), Progressively Worsening Group (14.35%), Moderate Stability Group (31.90%) and Consistently Deteriorating Group (8.49%). Among the predictive models, the XGBoost model achieved the best performance, with an accuracy of 0.664 (95%CI: 0.648-0.681), a macro ROC-AUC of 0.825 (95%CI: 0.816-0.834), a micro ROC-AUC of 0.884 (95%CI: 0.876-0.892), and a log loss of 0.806 (95%CI: 0.781-0.831). Other models, including Random Forest, Support Vector Machine, Logistic Regression, and Artificial Neural Networks, showed similar accuracy and ROC-AUC values. The study identified three key factors—baseline disease counts, self-rated Activities of Daily Living (ADL), and self-rated health status—as critical predictors of multimorbidity trajectories.

Keywords Multimorbidity, Trajectory, Prediction model

With the acceleration of global population aging, the high prevalence and coexistence of chronic diseases have become a major challenge in the field of global health, collectively referred to as multimorbidity. Multimorbidity describes the presence of two or more chronic conditions in a single individual, underscoring the complexity and diversity of their health needs¹. The interactions among these conditions can exacerbate the overall disease burden², significantly affecting patients' overall health³, functional capacity⁴, and quality of life⁵. Additionally, patients with multimorbidity often face higher rates of hospitalization, readmission, and long-term care needs, leading to substantial healthcare costs and placing a heavy burden on healthcare systems worldwide^{6–8}. Zhai et al. (2023)² concluded that elderly patients with multimorbidity, particularly those with multi-system or cardiovascular conditions, face a significantly higher risk of catastrophic health expenditures compared to those with fewer chronic conditions, highlighting the substantial economic burden associated with managing multiple chronic diseases. Due to the complexity of their conditions, these patients often require frequent outpatient visits and multidisciplinary care, significantly increasing the consumption of healthcare resources, particularly in countries with rapidly aging populations⁹.

Although variations in definitions and measurement methods have led to discrepancies in the reported prevalence of multimorbidity¹⁰, numerous studies consistently indicate a high prevalence of multimorbidity

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globally, especially among older populations. For example, a large cross-sectional study in Europe reported a multimorbidity prevalence of 23.2% in the general population, with rates rising to 64.9% among individuals aged 65 and above. In the United States, a national survey showed that the prevalence of multimorbidity among those aged 30 and above reached 62.2%, with 34.2% having three or more chronic conditions¹¹. The situation in China is similarly concerning, with recent studies indicating that approximately 56.7% of individuals aged 45 and above experience multimorbidity⁷. Moreover, researchers have conducted systematic reviews of the literature and reached similar conclusions¹². These figures underscore that multimorbidity has become a critical challenge that demands urgent attention in global public health. In response to this growing concern, the early identification of high-risk individuals with multimorbidity and the implementation of dynamic management strategies are crucial for improving patients' quality of life, delaying disease progression, and reducing healthcare costs¹³.

However, most existing studies on multimorbidity are limited to static assessments at single time points, making it difficult to capture the complex interactions between chronic diseases and their evolving trajectories over time. In reality, the onset and progression of chronic conditions are dynamic and gradual processes, with significant variations in multimorbidity trajectories among individuals¹⁴. These variations are influenced not only by physiological and genetic factors but also by lifestyle and socioeconomic environments¹⁵. Therefore, exploring the heterogeneity of multimorbidity trajectories in the older population is essential. Identifying potential patterns is key to optimizing management strategies.

In recent years, with the rapid advancement of big data and artificial intelligence technologies, machine learning has provided new solutions for identifying and predicting multimorbidity trajectories¹⁶. Particularly in time-series analysis, machine learning can capture the underlying patterns of multimorbidity, identify patient groups with varying levels of risk, and predict their future health outcomes. These technological advantages make machine learning a powerful tool for investigating multimorbidity trajectories and their impact, providing robust support for personalized health management and precision medicine. Based on this, the study utilizes data from the China Health and Retirement Longitudinal Study (CHARLS) and employs time-series clustering algorithms and machine learning models to develop predictive models of multimorbidity trajectories in middle-aged and older adults. The study aims to uncover different patterns of multimorbidity progression across various groups, providing a scientific basis for developing tailored intervention strategies, thereby facilitating more precise and effective management of multimorbidity.

Methods

Study design and participants

The data used in this study came from the CHARLS database, a comprehensive national longitudinal survey designed to monitor the health status of middle-aged and older adults (mainly aged 45 and above) across 28 provinces in China¹⁷. The survey employed a multistage stratified probability sampling method proportional to size and conducted follow-up assessments every 2 to 3 years. To extend the study's time span, data from the first wave (2011) to the fifth wave (2020) were utilized. Participants were excluded if they met any of the following criteria: (1) baseline age under 45, (2) absence from two consecutive follow-up waves, or (3) incomplete information related to diseases. This resulted in a final sample of 12,198 participants. Detailed information regarding the follow-up data and the sample selection process is provided in eFigure 1 in Supplementary file 1. This study was approved by the Biomedical Ethics Committee of Peking University (IRB00001052-11,015). We adhered to the TRIPOD + AI checklist for the reporting of prediction model studies¹⁸ (Supplementary file 2).

Multimorbidity assessment

In this study, the Chinese Multimorbidity-Weighted Index (CMWI) was utilized to quantify multimorbidity status among middle-aged and elderly patients¹⁹. This index was developed based on the CHARLS database, taking into account the impact of multiple chronic conditions on physical functioning, and providing a more suitable method for evaluating multimorbidity in the Chinese middle-aged and elderly population. Compared to a simple disease count, CMWI more accurately reflects the differential impact of various diseases on patient health, and it includes 14 chronic conditions, each weighted according to its impact on physical functioning¹⁹. The final CMWI score is calculated as the sum of the weights of the diagnosed conditions for each patient. Among the 14 conditions included, the three with the highest weights are stroke (5.1), memory-related conditions (4.3), and cancer (3.4). The weights of other conditions are detailed in eTable 1 of Supplementary 1. This study calculated the CMWI score at each follow-up time point for middle-aged and elderly individuals. For patients lost to follow-up in intermediate years, their CMWI score from the previous follow-up was used for imputation. Notably, none of the study participants experienced a loss to follow-up between two consecutive time points.

Trajectory analysis

This study utilized time-series K-means clustering based on Dynamic Time Warping (DTW) to analyze the trajectories of patients' multimorbidity indices^{20,21}. This method is well-suited for handling time-series data, using DTW to measure the similarity between time series and cluster data with similar patterns of change into the same group. This approach can reveal the underlying structure and dynamic patterns of the changes in multimorbidity indices over time. We employed the time-series K-means algorithm with DTW as the distance metric to identify patterns in the changes of the multimorbidity indices over time. The clustering process involves randomly initializing cluster centroids, assigning data points based on DTW distances, recalculating cluster centroids, and monitoring the changes in the centroids to determine convergence. In our clustering analysis, we identified the optimal number of clusters (K) by jointly assessing three criteria—the elbow plot of the sum of squared errors (SSE), the silhouette coefficient (SC) curve, and the Krzanowski-Lai (KL) index—and selected the K that best captured the dominant structure of the data^{22,23}. We optimized the model using Grid Search. By training the model with different parameter values and evaluating the model performance using evaluation

metrics such as the SC, we selected the optimal parameter combination. This approach allows the optimization of the DTW window parameters, minimizing intra-cluster variance and improving the clustering performance of the model. For a comprehensive evaluation of the performance of the constructed trajectory clustering model for patients' multimorbidity indices, multiple standardized metrics were used for quantitative analysis. These included the adjusted Akaike Information Criterion (aAIC), adjusted Bayesian Information Criterion (aBIC), SSE, SC, average intra-cluster distance, average inter-cluster distance, Bouldin index (DB) and Variance Explained (VarExp)^{24–28}. aAIC and aBIC were used to evaluate model complexity and superiority, with lower values indicating better model balance, reducing information loss and preventing overfitting. SSE reflects the compactness of data points within clusters, with lower SSE values indicating more concentrated distributions, which aids in determining the optimal K. The SC evaluates the cohesion and separation of the clustering results, with values ranging from –1 to 1, where values closer to 1 indicate better intra-cluster cohesion and inter-cluster separation. The average intra-cluster distance and average inter-cluster distance further assess the structural characteristics of the clustering, where lower average intra-cluster distance indicates tighter clustering, and higher average inter-cluster distance indicates clearer distinctions between clusters. The DB index is the ratio of the within-cluster distance to the between-cluster distance, where a lower value indicates better performance (i.e., smaller distances within the clusters and larger distances between different clusters)²⁷. VarExp is another important metric to evaluate clustering results, representing the ratio of between-cluster variance to total variance²⁸. The higher this value, the better the clustering model explains the overall variability of the data.

Additionally, to ensure the practical applicability of the clustering results, we required each trajectory cluster category to include at least 5% of the total sample size, thereby avoiding statistical bias due to small sample sizes in certain categories. Furthermore, the cluster categories should possess clear clinical significance, facilitating interpretation and application in subsequent health management and intervention strategies.

Input variables

Based on previous literature and professional judgment, the input variables in this study were collected according to participants' individual characteristics, lifestyle, socioeconomic status, physical function, healthcare utilization, caregiving behavior, and self-assessed status in 2011. These variables were obtained through questionnaires and physical examinations, then appropriately categorized and coded. Detailed information on these variables can be found in Supplementary 1 (eTable 2). The specific variables included (1) Sociodemographic Variables: These encompassed age, gender, education level, marital status, and other relevant factors. (2) Lifestyle Variables: These pertained to smoking, alcohol consumption, dietary habits, and other behaviors. (3) Socioeconomic Status: This included employment status, housing situation, and other relevant factors. (4) Physical Function: Data were collected through physical examinations and interviews; the Activities of Daily Living (ADL) self-assessment comprised 20 tasks, such as "running or jogging about 1 km," "walking 1 km," "carrying weights over 10 jin," and "picking up a small coin from a table." Each task was coded as 1 ("no difficulty"), 2 ("some difficulty but can complete"), 3 ("difficulty, requires assistance"), or 4 ("unable to complete"). For analytical convenience, ADL self-rating scores were classified as follows: a total score of 20 was categorized as level 1, scores between 21 and 40 as level 2, and scores over 40 as level 3. Other physical function variables included Body Mass Index (BMI), walking speed, grip strength among others. (5) Healthcare Utilization: This encompasses family medical services, blood pressure monitoring, and regular physical examinations. (6) Caregiving Behavior: This included whether the participant provided care to grandchildren or elderly individuals in the past year. (7) Self-assessed Status: This was based on participants' self-assessment of life expectancy, life satisfaction and health status. To mitigate the impact of missing values on predictors in the analysis, a multiple imputation method (with five imputations) was employed. This approach was implemented to ensure robust handling of incomplete data, reducing potential biases and preserving the integrity of the predictive model²⁹.

Feature selection

In this study, the XGBoost algorithm was employed to perform feature selection from multiple input features. As a widely used Gradient Boosting Decision Tree (GBDT) model, XGBoost was capable of efficiently handling large-scale data and complex nonlinear problems³⁰. Its built-in feature importance scoring mechanism was used to effectively identify the most influential features for model prediction, facilitating targeted feature selection³¹. Initially, the XGBoost model was trained using all available features to compute the relative importance of each feature. The features were then ranked in descending order based on their importance scores. Subsequently, features were progressively introduced according to their rankings, and ten-fold cross-validation was used to evaluate the model's classification performance after each addition. By comparing the cross-validation results of various feature combinations, the feature set that yielded the highest multiclass ROC–AUC was selected as the optimal feature subset. Once the optimal feature set was determined, it was further utilized to train other machine learning models, such as Random Forest (RF) and Support Vector Machines (SVM).

Predictive model development and evaluation

After selecting the optimal feature variables, we employed five commonly used machine learning algorithms, i.e., XGBoost, RF, SVM, Logistic Regression (LR), and Artificial Neural Network (ANN)—to construct models for predicting the trajectories of multimorbidity in middle-aged and elderly people. XGBoost, as a gradient-boosting tree model, can handle nonlinear relationships between features and address the issue of class imbalance³⁰. Random Forest reduces model variance effectively through ensemble learning³². XGBoost focuses on making the final decision through strong trees, thereby minimizing bias, while RF focuses on majority voting across all trees, emphasizing the reduction of variance. SVM performs well in high-dimensional spaces and is suitable for complex boundary classification³². LR is simple and interpretable, ideal for linearly separable data³⁴, while ANN has powerful nonlinear fitting capabilities, making them suitable for capturing complex

feature relationships³⁵. The dataset was randomly split into a training set (75%) and a test set (25%). During model training, tenfold cross-validation was employed to ensure the generalization ability and stability of the models. Grid Search was used to find the best hyperparameter combinations to improve model performance. We selected multiple evaluation metrics to comprehensively assess the classification ability of the models, including Accuracy, Balanced Accuracy, Log Loss, Macro Precision, Macro F1, Macro ROC–AUC, Micro ROC–AUC, and MCC (Matthews Correlation Coefficient). These metrics provide a well-rounded evaluation of the models in the context of class imbalance³⁶. Specifically, Accuracy measures the overall correctness, while Balanced Accuracy evaluates the model's average performance by accounting for class imbalance. Log Loss measures the uncertainty of the model's predictions by penalizing false classifications more severely, offering a more granular view of model performance. Macro Precision and Macro F1 assess the precision, recall, and harmonic mean for each class. Macro ROC–AUC evaluates the model's discriminatory ability for each class, whereas Micro ROC–AUC considers the overall performance across all classes. MCC, which considers true and false positives and negatives, provides a balanced evaluation even when class distribution is imbalanced. Additionally, to visually present model performance, we plotted ROC–AUC curves and Precision–Recall curves. To enhance the interpretability of the models, we applied SHapley Additive exPlanations (SHAP) values for feature importance analysis.

Statistical analysis

The primary statistical analysis comprised three key steps. In the first step, the time-series K-means algorithm was utilized to conduct a clustering analysis on the trajectories of patients with multimorbidity. The second step involved the application of the XGBoost model for feature selection. In the third step, the optimally selected features were employed across five machine learning models to predict the trajectories of multimorbidity. SHAP analysis was also conducted to explain significant factors influencing various multimorbidity trajectory patterns. Continuous variables were reported as mean $\pm$ standard deviation, whereas categorical variables were expressed as percentages. The baseline characteristics among different trajectory groups were compared using one-way analysis of variance (ANOVA) for continuous variables, rank-sum test for ordinal categorical variables, and the chi-square test for categorical variables. A two-sided P-value of less than 0.05 was deemed statistically significant. All analyses were performed using Python 3.11, incorporating major libraries such as Scikit-learn, XGBoost, SHAP, Pandas, and NumPy.

Results

Heterogeneous trajectories of multimorbidity

The optimal number of clusters was determined to be four (eFigure 2 in Supplementary file 1). When the number of clusters increased further, the reduction in SSE became marginal, and the SC and KL indices were highest at $K = 4$, indicating that the four-cluster solution strikes an appropriate balance between within-cluster compactness and between-cluster separation. Additionally, metrics such as an aAIC of $-483,884.152$, an aBIC of $-483,706.336$, a SC of 0.380, an average intra-cluster distance of 1.669, an average inter-cluster distance of 3.276, a DB of 1.015 and VarExp 0.703, further support the superiority of the four-cluster solution in terms of explanatory power and distinctiveness. The time-series clustering results categorize the multimorbidity progression trajectories into four groups (Fig. 1) as follows: (1) Stable Low-Risk Group (green): comprising 5521 individuals, accounting for 45.26%; (2) Progressively Worsening Group (blue): comprising 1750 individuals, accounting for 14.35%; (3) Moderate Stability Group (orange): comprising 3891 individuals, accounting for 31.90%; and (4) Consistently Deteriorating Group (red): comprising 1036 individuals, accounting for 8.49%. Each group exhibits significantly different trends in CMWI scores over time, reflecting the heterogeneity in disease progression among patients with multimorbidity.

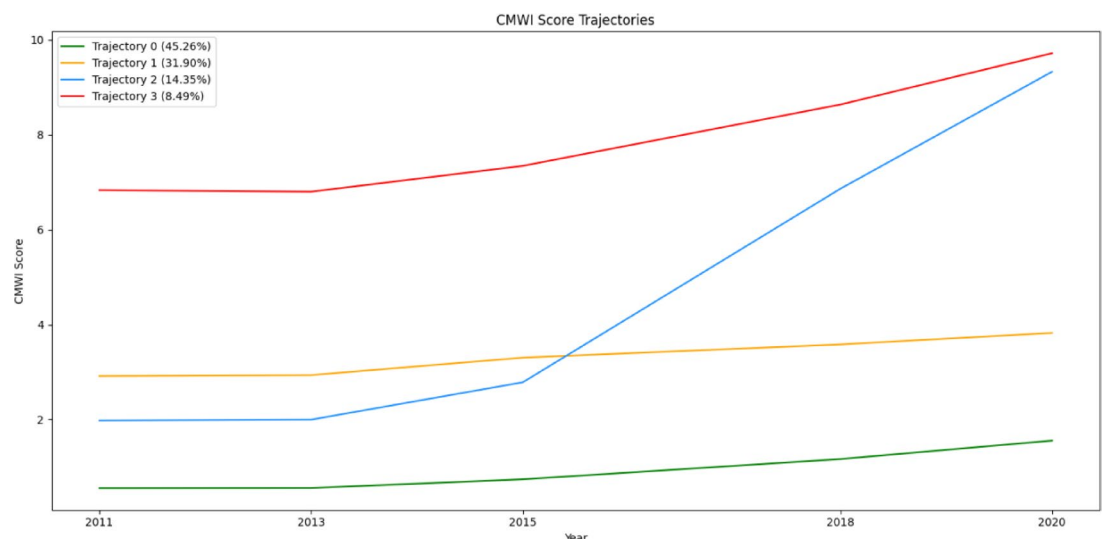


Fig. 1. Heterogeneous trajectory cluster of multimorbidity.

Baseline characteristics of study population

Table 1 presents the baseline characteristics of study participants across different trajectory categories. As previously mentioned, this study identified 5521 participants in the Stable Low-Risk Group, 1750 in the Progressively Worsening Group, 3891 in the Moderate Stability Group, and 1036 in the Consistently Deteriorating Group. Significant differences were observed among the groups in various characteristics, including age, gender, education level, marital status, number of children, smoking, and alcohol consumption ($p < 0.05$). More detailed information can be found in Table 1. In terms of age distribution, the highest proportion of individuals aged 45–59 was found in the Stable Low-Risk Group (69.4%), while the Consistently Deteriorating Group had the largest proportion of participants aged 60–79 (54.1%). In gender distribution, the proportion of females was highest in the Consistently Deteriorating Group (58.3%). Regarding educational attainment, the illiteracy rate was highest in the Progressively Worsening Group (29.8%). Marital status differed across groups, with the Stable Low-Risk Group showing the highest rate of marriage (92.3%), while the Consistently Deteriorating Group had a higher proportion of divorced, widowed, or single individuals (13.9%). Smoking and alcohol consumption behaviors also varied, with the Stable Low-Risk Group having the highest proportions of individuals who did not smoke or drink alcohol (60.0% and 87.7%, respectively). In terms of health-related characteristics, the Consistently Deteriorating Group reported a poorer self-rated health status, with the proportions of individuals rating their health as "poor" and "very poor" being significantly higher than those in other groups (55.6% and 7.8%, respectively). Furthermore, the Consistently Deteriorating Group demonstrated a higher total number of diseases (4.05 ± 1.25), a greater history of falls (27.0%), and a larger proportion of individuals with disabilities (7.3%) compared to other groups. Overall, significant differences in various baseline characteristics across groups highlight the heterogeneity among patients with multimorbidity in terms of sociodemographic factors, lifestyle choices, and health status. These differences may have implications for the distinct trajectories of disease progression. Additionally, to more precisely explain the factors influencing multimorbidity progression, we focused on the characteristic variables of the Progressively Worsening Group. The results showed significant differences between the Progressively Worsening Group and non-Progressively Worsening Groups in terms of age, depressive symptoms, self-rated health status, and other factors. Further details can be found in Supplementary file 1, eTable 3.

Predictive model development and evaluation

Using the XGBoost algorithm, we selected the three key variables that optimized the model's performance: self-rated health status, ADL self-rating, and baseline disease counts. The feature importance ranking is shown in eFigure 3A of Supplementary file 1, with cumulative feature importance presented in eFigure 4, and model performance with varying numbers of features is shown in eFigure 5. Additionally, to better understand the impact of different features on multimorbidity progression trajectories, we compared feature importance between the XGBoost and RF models and created a feature importance plot based on the RF algorithm (see eFigure 3B) for a detailed comparison. Subsequently, we employed five machine learning models to predict multimorbidity progression trajectories in middle-aged and older adults, as shown in Table 2. The XGBoost model performed well on several metrics, including accuracy (0.664, 95% CI: 0.648–0.681), macro ROC-AUC (0.825, 95% CI: 0.816–0.834), micro ROC-AUC (0.884, 95% CI: 0.876–0.892), and log loss (0.806, 95% CI: 0.781–0.831). The RF and ANN models showed similar performance, achieving micro ROC-AUC values of 0.883 and 0.884, respectively, and log loss values of 0.821 and 0.806. The LR model achieved the highest macro precision score of 0.721 (95% CI: 0.498–0.772), indicating an advantage in precision. The SVM model performed slightly less favorably in terms of accuracy and ROC-AUC, with a macro ROC-AUC of 0.795 (95% CI: 0.785–0.805) and micro ROC-AUC of 0.864 (95% CI: 0.855–0.872); however, it still maintained stable performance in terms of log loss and MCC. The macro ROC curves for the different models are presented in Fig. 2, the macro precision-recall curves in Fig. 3, and the micro ROC and micro precision-recall curves in eFigure 6 and eFigure 7 (Supplementary file 1), respectively. Using the XGBoost model as an example, the macro ROC curves and macro precision-recall curves for each category are shown in Supplementary file 1, eFigure 8 and eFigure 9, respectively. These evaluation results indicate that machine learning methods such as XGBoost, RF, and ANN all have good applicability in predicting multimorbidity trajectories, with each model demonstrating advantages across different performance metrics.

Interpretability analysis

In the model interpretability analysis, we utilized XGBoost and RF models, combined with SHAP analysis, to quantitatively evaluate the influence of different variables on the prediction outcomes. Through SHAP value distribution plots (Fig. 4), we were able to clearly understand the direction and magnitude of each feature's impact on the predictions. It can be observed that the three optimal model features selected by the XGBoost model—baseline disease counts, self-rated health status, and ADL self-rated score—are key predictive variables for the multimorbidity trajectory. Furthermore, to further analyze the specific roles of each feature in different trajectory categories, we also plotted the SHAP value distribution for the four trajectory categories (eFigure 10 and eFigure 11, see Supplementary File 1 for details). Overall, in both the XGBoost and RF models, the baseline disease counts, self-rated health status, and ADL self-rating consistently demonstrated similar roles across all trajectory categories, with baseline disease counts and health status showing a more significant impact. Specifically, fewer diseases, better health status (i.e., lower scores), and higher self-care ability (i.e., lower ADL scores) contributed to predicting the Stable Low-Risk Group. When health status was poorer (i.e., higher scores), individuals were more likely to be predicted as part of the Progressively Worsening Group. When disease count increased and health status worsened (i.e., higher scores), individuals were more likely to be predicted as part of the Progressively Worsening Group, with the effect of disease count showing a fluctuating trend, indicating the instability of baseline disease count in this group. A higher disease count, poorer health status (i.e., higher scores),

Variables	Stable low-risk group (5521, 45.26%)	Moderate stability group (3891, 31.90%)	Progressively worsening group (1750, 14.35%)	Consistently deteriorating group (1036, 8.49%)	$X^2/F/W$	P
Age						
45–59	3831 (69.4)	2314 (59.5)	894 (51.1)	449 (43.3)	374.028	<0.001
60–79	1623 (29.4)	1526 (39.2)	824 (47.1)	560 (54.1)		
≥80	67 (1.2)	51 (1.3)	32 (1.8)	27 (2.6)		
Gender						
Male	2812 (50.9)	1691 (43.5)	801 (45.8)	432 (41.7)	66.605	<0.001
Female	2709 (49.1)	2200 (56.5)	949 (54.2)	604 (58.3)		
Education						
Illiterate	1259 (22.8)	1086 (27.9)	521 (29.8)	308 (29.7)	157.385	<0.001
Elementary school and below	865 (15.7)	764 (19.6)	317 (18.1)	217 (20.9)		
Elementary school	1185 (21.5)	858 (22.1)	426 (24.3)	233 (22.5)		
Junior high school	1391 (25.2)	787 (20.2)	305 (17.4)	174 (16.8)		
High school	563 (10.2)	251 (6.5)	116 (6.6)	59 (5.7)		
College and above	258 (4.7)	145 (3.7)	65 (3.7)	45 (4.3)		
Marital status						
Married	5098 (92.3)	3449 (88.6)	1552 (88.7)	892 (86.1)	73.001	<0.001
Separated/divorced	63 (1.1)	42 (1.1)	21 (1.2)	15 (1.4)		
Widowed	321 (5.8)	372 (9.6)	165 (9.4)	118 (11.4)		
Single	39 (0.7)	28 (0.7)	12 (0.7)	11 (1.1)		
Adult school						
No	5215 (94.5)	3663 (94.1)	1668 (95.3)	967 (93.3)	5.481	0.140
Yes	306 (5.5)	228 (5.9)	82 (4.7)	69 (6.7)		
Residence						
Rural	3641 (65.9)	2644 (68.0)	1181 (67.5)	685 (66.1)	4.719	0.194
Urban	1880 (34.1)	1247 (32.0)	569 (32.5)	351 (33.9)		
Medical insurance						
No	320 (5.8)	205 (5.3)	98 (5.6)	41 (4.0)	6.092	0.107
Yes	5201 (94.2)	3686 (94.7)	1652 (94.4)	995 (96.0)		
Number of children						
0	1270 (23.0)	834 (21.4)	337 (19.3)	220 (21.2)	9.110	0.028
1–2	1739 (31.5)	1307 (33.6)	573 (32.7)	314 (30.3)		
3–4	1679 (30.4)	1156 (29.7)	557 (31.8)	336 (32.4)		
≥5	833 (15.1)	594 (15.3)	283 (16.2)	166 (16.0)		
Smoking						
No	3315 (60.0)	2468 (63.4)	1090 (62.3)	639 (61.7)	11.502	0.009
Yes	2206 (40.0)	1423 (36.6)	660 (37.7)	397 (38.3)		
Alcohol consumption						
No	4844 (87.7)	3347 (86.0)	1479 (84.5)	830 (80.1)	53.823	<0.001
< 1/month	458 (8.3)	309 (7.9)	157 (9.0)	90 (8.7)		
> 1/month	219 (4.0)	235 (6.0)	114 (6.5)	116 (11.2)		
Smoking index						
No smoking	3688 (66.8)	2664 (68.5)	1176 (67.2)	689 (66.5)	3.070	0.381
1–100	107 (1.9)	56 (1.4)	25 (1.4)	19 (1.8)		
100–200	100 (1.8)	77 (2.0)	36 (2.1)	20 (1.9)		
> 200	1626 (29.5)	1094 (28.1)	513 (29.3)	308 (29.7)		
Regular diet						
No	581 (10.5)	560 (14.4)	239 (13.7)	162 (15.6)	43.279	<0.001
Yes	4940 (89.5)	3331 (85.6)	1511 (86.3)	874 (84.4)		
Bedtime (h)						
< 5	544 (9.9)	674 (17.3)	275 (15.7)	267 (25.8)	229.543	<0.001
5–8	2772 (50.2)	1966 (50.5)	908 (51.9)	508 (49.0)		
≥8	2205 (39.9)	1251 (32.2)	567 (32.4)	261 (25.2)		
Nap (min)						
0	2882 (52.2)	1998 (51.3)	900 (51.4)	497 (48.0)	4.289	0.232
1–30	463 (8.4)	328 (8.4)	129 (7.4)	105 (10.1)		
Continued						

Variables	Stable low-risk group (5521, 45.26%)	Moderate stability group (3891, 31.90%)	Progressively worsening group (1750, 14.35%)	Consistently deteriorating group (1036, 8.49%)	$X^2/F/W$	P
30 ~ 60	1286 (23.3)	946 (24.3)	423 (24.2)	261 (25.2)		
60 ~ 120	833 (15.1)	572 (14.7)	281 (16.1)	161 (15.5)		
≥ 120	57 (1.0)	47 (1.2)	17 (1.0)	12 (1.2)		
Employment status						
No	4487 (81.3)	3431 (88.2)	1547 (88.4)	962 (92.9)	159.061	<0.001
Yes	1034 (18.7)	460 (11.8)	203 (11.6)	74 (7.1)		
Salary						
No	4153 (75.2)	3279 (84.3)	1526 (87.2)	933 (90.1)	246.575	<0.001
Yes	1368 (24.8)	612 (15.7)	224 (12.8)	103 (9.9)		
Renting						
No	5441 (98.6)	3822 (98.2)	1727 (98.7)	1017 (98.2)	2.777	0.427
Yes	80 (1.4)	69 (1.8)	23 (1.3)	19 (1.8)		
Social score						
0	2901 (52.5)	2054 (52.8)	973 (55.6)	551 (53.2)	6.893	0.075
1 ~ 5	1845 (33.4)	1350 (34.7)	571 (32.6)	358 (34.6)		
≥ 5	775 (14.0)	487 (12.5)	206 (11.8)	127 (12.3)		
Mortgaging a house						
No	5427 (98.3)	3812 (98.0)	1724 (98.5)	1014 (97.9)	3.027	0.388
Yes	94 (1.7)	79 (2.0)	26 (1.5)	22 (2.1)		
ADL self-rating						
20	2770 (50.2)	1037 (26.7)	419 (23.9)	83 (8.0)	1205.791	<0.001
21 ~ 40	2702 (48.9)	2749 (70.7)	1267 (72.4)	849 (81.9)		
> 40	49 (0.9)	105 (2.7)	64 (3.7)	104 (10.0)		
BMI (kg/m ²)						
< 18.5	240 (4.3)	221 (5.7)	78 (4.5)	48 (4.6)	130.285	<0.001
18.5 ~ 23.9	3694 (66.9)	2320 (59.6)	1012 (57.8)	521 (50.3)		
24 ~ 27.9	1227 (22.2)	941 (24.2)	450 (25.7)	295 (28.5)		
≥ 28	360 (6.5)	409 (10.5)	210 (12.0)	172 (16.6)		
Walking speed time (s)						
< 3	4442 (80.5)	2760 (70.9)	1174 (67.1)	621 (59.9)	290.418	<0.001
3 ~ 10	1058 (19.2)	1096 (28.2)	554 (31.7)	398 (38.4)		
> 10	21 (0.4)	35 (0.9)	22 (1.3)	17 (1.6)		
Left chest pain						
No	5013 (90.8)	3117 (80.1)	1431 (81.8)	632 (61.0)	632.209	<0.001
Yes	508 (9.2)	774 (19.9)	319 (18.2)	404 (39.0)		
Disability						
No	5359 (97.1)	3731 (95.9)	1682 (96.1)	960 (92.7)	47.566	<0.001
Yes	162 (2.9)	160 (4.1)	68 (3.9)	76 (7.3)		
Distance eyesight						
Excellent	104 (1.9)	46 (1.2)	21 (1.2)	6 (0.6)	424.734	<0.001
Very good	847 (15.3)	374 (9.6)	175 (10.0)	60 (5.8)		
Good	1478 (26.8)	855 (22.0)	371 (21.2)	157 (15.2)		
Fair	2350 (42.6)	1749 (44.9)	796 (45.5)	448 (43.2)		
Poor	742 (13.4)	867 (22.3)	387 (22.1)	365 (35.2)		
Near eyesight						
Excellent	78 (1.4)	35 (0.9)	12 (0.7)	6 (0.6)	183.697	<0.001
Very good	567 (10.3)	273 (7.0)	137 (7.8)	45 (4.3)		
Good	1337 (24.2)	837 (21.5)	357 (20.4)	171 (16.5)		
Fair	2589 (46.9)	1820 (46.8)	823 (47.0)	500 (48.3)		
Poor	950 (17.2)	926 (23.8)	421 (24.1)	314 (30.3)		
Hearing ability						
Excellent	104 (1.9)	41 (1.1)	17 (1.0)	7 (0.7)	470.552	<0.001
Very good	1015 (18.4)	479 (12.3)	205 (11.7)	68 (6.6)		
Good	1950 (35.3)	1120 (28.8)	471 (26.9)	203 (19.6)		
Fair	1971 (35.7)	1729 (44.4)	797 (45.5)	517 (49.9)		
Continued						

Variables	Stable low-risk group (5521, 45.26%)	Moderate stability group (3891, 31.90%)	Progressively worsening group (1750, 14.35%)	Consistently deteriorating group (1036, 8.49%)	$\chi^2/F/W$	P
Poor	481 (8.7)	522 (13.4)	260 (14.9)	241 (23.3)		
Weight change						
No change	5056 (91.6)	3420 (87.9)	1538 (87.9)	831 (80.2)	137.081	<0.001
Only weight gain	131 (2.4)	141 (3.6)	55 (3.1)	51 (4.9)		
Only weight loss	267 (4.8)	278 (7.1)	138 (7.9)	132 (12.7)		
Gained weight first, then lost weight	36 (0.7)	24 (0.6)	11 (0.6)	14 (1.4)		
Lost weight first, then gained weight	31 (0.6)	28 (0.7)	8 (0.5)	8 (0.8)		
Activity score						
< 16	791 (14.3)	614 (15.8)	273 (15.6)	191 (18.4)	10.796	0.013
16–17	3575 (64.8)	2481 (63.8)	1153 (65.9)	644 (62.2)		
> 17	1155 (20.9)	796 (20.5)	324 (18.5)	201 (19.4)		
Sit-to-stand test						
Fail	39 (0.7)	48 (1.2)	36 (2.1)	28 (2.7)	40.547	<0.001
Success	5482 (99.3)	3843 (98.8)	1714 (97.9)	1008 (97.3)		
Grip strength						
Normal	5005 (90.7)	3347 (86.0)	1504 (85.9)	827 (79.8)	118.246	<0.001
Decreased	516 (9.3)	544 (14.0)	246 (14.1)	209 (20.2)		
Health before 15 years old						
Excellent	549 (9.9)	344 (8.8)	150 (8.6)	92 (8.9)	76.124	<0.001
Very good	2273 (41.2)	1397 (35.9)	688 (39.3)	349 (33.7)		
Good	1534 (27.8)	1087 (27.9)	471 (26.9)	263 (25.4)		
Fair	873 (15.8)	754 (19.4)	323 (18.5)	218 (21.0)		
Poor	292 (5.3)	309 (7.9)	118 (6.7)	114 (11.0)		
Cognitive impairment						
No	4583 (83.0)	3242 (83.3)	1438 (82.2)	850 (82.0)	1.721	0.632
Yes	938 (17.0)	649 (16.7)	312 (17.8)	186 (18.0)		
Teeth loss						
No	5234 (94.8)	3606 (92.7)	1613 (92.2)	932 (90.0)	44.575	<0.001
Yes	287 (5.2)	285 (7.3)	137 (7.8)	104 (10.0)		
Depressive symptoms						
No	4085 (74.0)	2265 (58.2)	976 (55.8)	433 (41.8)	565.416	<0.001
Yes	1436 (26.0)	1626 (41.8)	774 (44.2)	603 (58.2)		
Bedridden status						
No	5510 (99.8)	3867 (99.4)	1732 (99.0)	1014 (97.9)	59.098	<0.001
Yes	11 (0.2)	24 (0.6)	18 (1.0)	22 (2.1)		
Disease counts	0.51 ± 0.64	1.48 ± 0.99	1.94 ± 0.94	4.05 ± 1.25	5755.052	<0.001
Fall						
No	4953 (89.7)	3247 (83.4)	1477 (84.4)	756 (73.0)	225.437	<0.001
Yes	568 (10.3)	644 (16.6)	273 (15.6)	280 (27.0)		
Executive function						
Normal	2045 (37.0)	1631 (41.9)	742 (42.4)	488 (47.1)	51.757	<0.001
Decreased	3476 (63.0)	2260 (58.1)	1008 (57.6)	548 (52.9)		
Home medical service						
No	5493 (99.5)	3853 (99.0)	1735 (99.1)	1018 (98.3)	18.749	0.001
Yes	28 (0.5)	38 (1.0)	15 (0.9)	18 (1.7)		
Blood pressure measurements						
0	5071 (91.8)	2921 (75.1)	1257 (71.8)	523 (50.5)	1198.663	<0.001
1 ~ 10	290 (5.3)	622 (16.0)	280 (16.0)	288 (27.8)		
11 ~ 20	82 (1.5)	176 (4.5)	109 (6.2)	94 (9.1)		
> 20	78 (1.4)	172 (4.4)	104 (5.9)	131 (12.6)		
Physical exam interval(years)						
Never	2131 (38.6)	1380 (35.5)	567 (32.4)	295 (28.5)	22.930	<0.001
≤ 1	2318 (42.0)	1808 (46.5)	823 (47.0)	560 (54.1)		
1 ~ 10	464 (8.4)	328 (8.4)	167 (9.5)	167 (9.5)		
> 10	608 (11.0)	375 (9.6)	193 (11.0)	72 (6.9)		
Continued						

Variables	Stable low-risk group (5521, 45.26%)	Moderate stability group (3891, 31.90%)	Progressively worsening group (1750, 14.35%)	Consistently deteriorating group (1036, 8.49%)	X ² /F/W	P
Take care of grandchildren						
No	4582 (83.0)	3092 (79.5)	1429 (81.7)	828 (79.9)	20.443	<0.001
Yes	939 (17.0)	799 (20.5)	321 (18.3)	208 (20.1)		
Take care of the elders						
No	5036 (91.2)	3564 (91.6)	1658 (94.7)	948 (91.5)	23.074	<0.001
Yes	485 (8.8)	327 (8.4)	92 (5.3)	88 (8.5)		
Life expectancy						
Almost impossible	176 (3.2)	269 (6.9)	161 (9.2)	156 (15.1)	574.043	<0.001
Not very likely	595 (10.8)	743 (19.1)	335 (19.1)	297 (28.7)		
Maybe	2255 (40.8)	1563 (40.2)	679 (38.8)	363 (35.0)		
Very likely	1432 (25.9)	838 (21.5)	367 (21.0)	142 (13.7)		
Almost certain	1063 (19.3)	478 (12.3)	208 (11.9)	78 (7.5)		
Life satisfaction						
Completely satisfied	61 (1.1)	94 (2.4)	37 (2.1)	58 (5.6)	89.469	<0.001
Very satisfied	528 (9.6)	458 (11.8)	236 (13.5)	154 (14.9)		
Somewhat satisfied	3803 (68.9)	2678 (68.8)	1186 (67.8)	663 (64.0)		
Not very satisfied	1030 (18.7)	617 (15.9)	617 (15.9)	146 (14.1)		
Not at all satisfied	99 (1.8)	44 (1.1)	28 (1.6)	15 (1.4)		
Health status						
Very good	735 (13.3)	161 (4.1)	83 (4.7)	9 (0.9)	1678.685	<0.001
Good	1351 (24.5)	472 (12.1)	221 (12.6)	39 (3.8)		
Fair	2703 (49.0)	1989 (51.1)	804 (45.9)	331 (31.9)		
Poor	691 (12.5)	1172 (30.1)	580 (33.1)	576 (55.6)		
Very poor	41 (0.7)	97 (2.5)	62 (3.5)	81 (7.8)		

Table 1. Sample characteristics by multimorbidity trajectories categories in 2011.

Models	Accuracy	Balanced accuracy	Macro precision	Macro F1	Macro roc auc	Micro roc auc	Log loss	Mcc
XGBoost	0.664(0.648–0–681)	0.543(0.525–0.560)	0.499(0.483–0.517)	0.517(0.501–0.533)	0.825(0.816–0.834)	0.884(0.876–0.892)	0.806(0.781–0.831)	0.478(0.454–0.502)
RF	0.665(0.649–0.682)	0.540(0.522–0.557)	0.500(0.484–0.517)	0.516(0.500–0.532)	0.823(0.813–0.832)	0.883(0.875–0.890)	0.821(0.789–0.859)	0.478(0.456–0.501)
SVM	0.666(0.650–0.682)	0.540(0.523–0.557)	0.500(0.484–0.517)	0.515(0.500–0.532)	0.795(0.785–0.805)	0.864(0.855–0.872)	0.874(0.847–0.899)	0.480(0.457–0.503)
LR	0.668(0.652–0.684)	0.539(0.522–0.557)	0.721(0.498–0.772)	0.519(0.503–0.535)	0.822(0.813–0.831)	0.882(0.875–0.890)	0.810(0.785–0.835)	0.482(0.460–0.505)
ANN	0.667(0.651–0.683)	0.541(0.523–0.558)	0.500(0.484–0.517)	0.516(0.500–0.532)	0.825(0.815–0.834)	0.884(0.876–0.891)	0.806(0.781–0.831)	0.481(0.458–0.503)

Table 2. Performance comparison of different machine learning models for multimorbidity trajectory prediction.

and lower self-care ability (i.e., higher ADL scores) were closely associated with predictions for the Consistently Deteriorating Group. For the Moderate Stability Group, the influence of the three variables showed a fluctuating pattern, indicating complex interactions among the variables. In addition to the key variables selected by the XGBoost model, we also conducted SHAP analysis on the top 20 most important features, quantifying their influence on the model's prediction results. Although these features were not fully included in the final feature set selected by XGBoost, the SHAP analysis provided insights into their specific roles in the multimorbidity trajectory, helping us understand other potential influencing factors. For further details, see eFigure 12 and eFigure 13 in Supplementary File 1.

Discussion

This study utilized data from the CHARLS and employed time-series clustering and machine learning methods to identify heterogeneous trajectories in the progression of multimorbidity among middle-aged and elderly individuals, and constructed a predictive model. The results reveal four distinct multimorbidity trajectories: a Stable Low-Risk Group, a Progressively Worsening Group, a Moderate Stability Group, and a Consistently Deteriorating Group. These findings are similar to those of Newman et al.³⁷. These trajectories not only reflect varying patterns in the progression of multimorbidity but also underscore the long-term health impact of disease

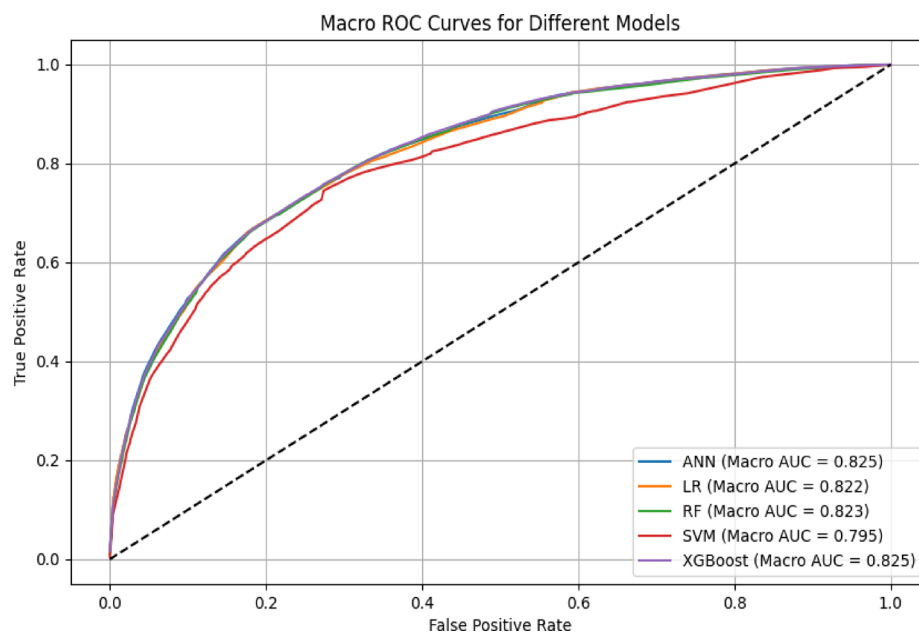


Fig. 2. Macro ROC curves for different models.

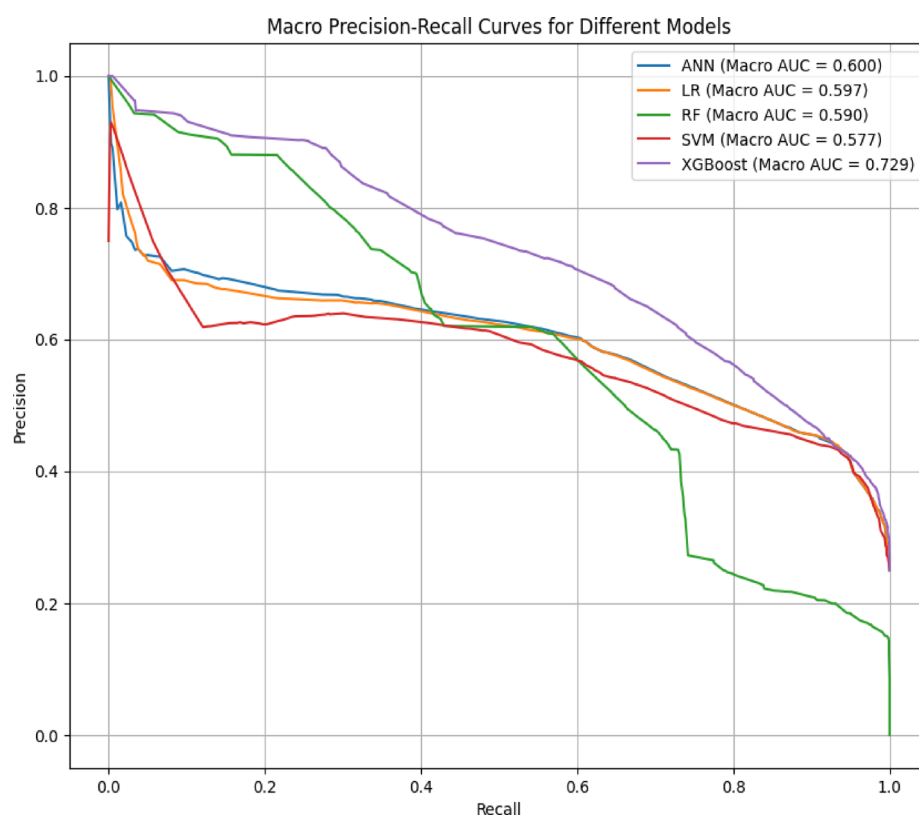


Fig. 3. Macro precision-recall curves for different models.

burden across different population groups. The classification of these groups shows significant differences in baseline characteristics, health status, and lifestyle across multiple dimensions. For instance, individuals in the Stable Low-Risk Group are predominantly middle-aged (45–59 years old), while those in the Consistently Deteriorating Group are mainly aged between 60 and 79 years. This may be attributed to the decline in immune and metabolic functions associated with aging, which typically facilitates the coexistence and worsening of chronic diseases¹⁵.

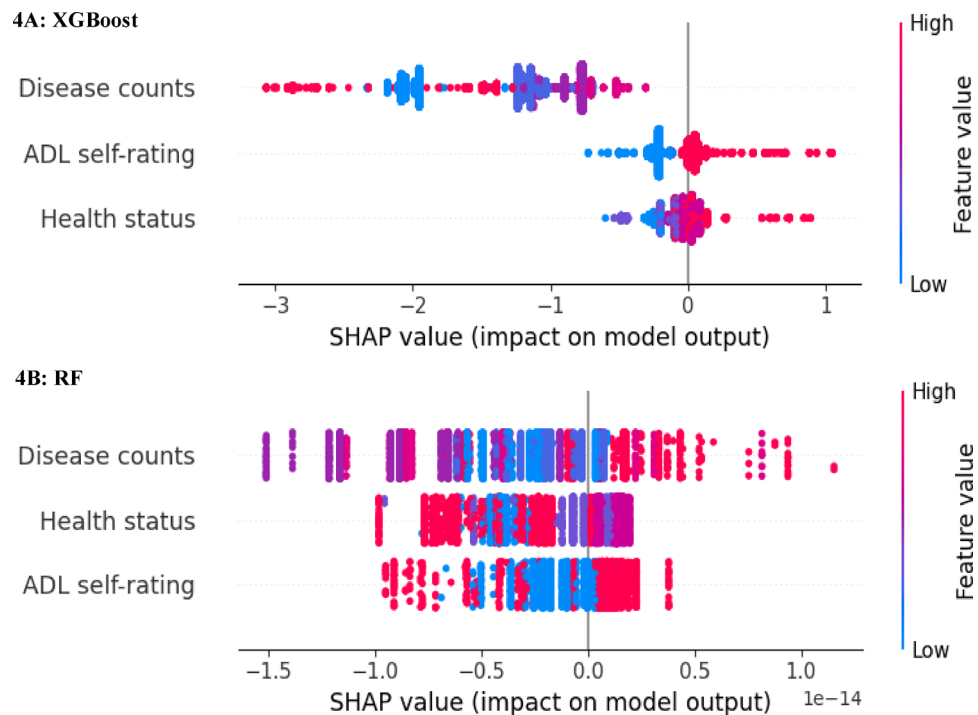


Fig. 4. SHAP summary plot of key features in XGBoost and RF models.

To further understand the predictive factors of multimorbidity progression trajectories, we conducted a comparative analysis using five machine learning models. The results show that the XGBoost model performed well across multiple metrics, including accuracy, ROC-AUC and log loss, particularly excelling with a micro-ROC-AUC of 0.884, demonstrating its strong predictive capability. Simultaneously, models such as RF and ANN also showed good performance. This indicates that machine learning methods hold significant practical value in predicting the progression of multimorbidity. Additionally, systematic reviews by Alsaleh et al.^{16,38,39} highlighted the promising potential of machine learning in multimorbidity research, and this study provides strong support for further research in this area.

In the model's interpretability analysis, baseline disease counts, ADL self-rating score, and self-rated health status were identified as key factors in predicting the progression of multimorbidity. This suggests that, in chronic disease management, these variables should be prioritized, particularly the ADL self-rating score and self-rated health status, as they are crucial in predicting high-risk trajectories. The findings of this study align with existing research on the heterogeneity and key factors influencing the progression of multimorbidity. For instance, Cezard et al. (2021) conducted a systematic review of multimorbidity trajectories, highlighting significant variations in disease burden among individuals¹⁴. However, many prior studies have concentrated on static cross-sectional analyses, neglecting the dynamic nature of multimorbidity progression¹⁰. By employing time-series clustering and machine learning techniques, this study supplements existing research by addressing the limitations in capturing the temporal dynamics of disease progression. Furthermore, this research utilizes the CMWI to quantify disease burden, thereby enhancing its applicability for assessing the health status of middle-aged and elderly patients with multimorbidity in China. The findings provide localized insights for managing chronic diseases within the context of an aging Chinese population and offer scientific evidence to guide the development of targeted intervention strategies and resource allocation^{40,41}.

From both biological and behavioral perspectives, the heterogeneity in multimorbidity progression may be influenced by multiple factors. Health status, disease counts, and ADL score were identified as critical influencing factors in this study. An increase in disease count not only directly intensifies individual disease burden but may also exacerbate disease progression through interactive effects¹³. Additionally, decreased self-care ability significantly increases the likelihood of high-risk trajectories, closely related to diminished daily functioning and greater social support needs^{42,43}. The findings of this study support these observations, suggesting that in multimorbidity management, attention should extend beyond the diseases themselves to include patients' functional capacity, allowing for the design of more personalized intervention strategies. Additionally, focusing on patients' self-rated health status is equally important. Self-rated health status, as a subjective evaluation of health by the patient, can reflect their overall health level, perceived disease burden, and the challenges they face in daily life. Research has shown that patients' self-reported outcomes are of significant value in health management and the assessment of treatment effectiveness⁴⁴. A patient-centered health assessment and management approach, which emphasizes patient involvement throughout care, not only better meets patients' needs but also improves treatment adherence. This, to some extent, can promote the implementation of more

precise, personalized medical plans, thereby better addressing the health management needs of multimorbid patients in an aging society^{45–47}.

This study holds significant implications for public health practice. First, identifying different trajectories of multimorbidity progression facilitates personalized management strategies for distinct risk groups in clinical practice. For patients in the “Progressively Worsening” and “Consistently Deteriorating” groups, early intervention may delay disease progression and reduce healthcare costs. Second, feature selection results demonstrate that self-rated health status, baseline disease counts, and ADL self-rating score possess strong predictive value for multimorbidity trajectories. These variables are not only easy to obtain but also support the rapid identification of high-risk patients in clinical settings. Furthermore, the high interpretability of the models enables clinicians to better understand the influence of different variables on multimorbidity trajectories, thus advancing the implementation of precision medicine and personalized health management.

Limitations

Although this study provides new insights into the identification and prediction of multimorbidity trajectories, several limitations persist. First, because the study relies on CHARLS data, its generalizability may be constrained by the specific characteristics of this dataset, potentially limiting its applicability to other populations. Second, while the CMWI index is suitable for the Chinese population, it may necessitate adaptation for multimorbidity research in other countries or regions. Third, missing values in the time-series data were imputed using information from the previous follow-up wave, which may overlook individual fluctuations at specific time points. Future research could investigate more precise imputation methods to further enhance the model's accuracy and robustness.

Conclusions

This study employed time-series clustering and machine learning methods to provide new insights into the progression trajectories of multimorbidity among middle-aged and elderly individuals. Four distinct disease progression patterns were identified, and the influence of key predictive factors was analyzed in-depth using the XGBoost algorithm, thereby offering a scientific foundation for personalized management of chronic diseases. These findings not only contribute to the optimization of clinical intervention strategies but also offer essential empirical support for addressing the challenges associated with multimorbidity management in the context of an aging population and the advancement of precision health management.

Data availability

The research data are sourced from the China Health and Retirement Longitudinal Study (CHARLS) database. The dataset can be obtained through the official CHARLS website at: (<http://www.charls.cn/>).

Code availability

The code used in this study is available upon reasonable request to the corresponding author.

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Conceptualization: L.Y. and Qiao.L.; Methodology: L.Y., Qiao.L., L.X. and D.Y.; Data curation: Z.Z., J.Y., T.W., Y.L. and Qin.L.; Software: L.Y., L.X. and D.Y.; Writing (original draft preparation): L.Y.; Writing (review and editing): L.Y., Qiao.L., Z.Z., J.Y., T.W., Y.L. and Qin.L.; Project administration: Qiao.L.; Funding acquisition: L.Y. and Qiao.L. All authors have read and agreed to the published version of the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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